Sparsity's Hidden Cost: Selection-Set Overfitting and Lost Predictive Power in Sparse Autoencoder Pose Filters

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Sparse autoencoders (SAEs) are increasingly used to decompose biological foundation models into interpretable features. This carries an implicit assumption: since reconstruction fidelity can be pushed arbitrarily high, a sparse representation should be a safe stand-in for the dense representation it was trained on. We test this assumption in a concrete, practically motivated drug-discovery setting—filtering poor docking poses from a physics-informed protein-ligand docking pipeline's variational autoencoder latent space, across a sweep of sparsity levels—and find that it does not hold. Dense latents consistently match or outperform sparse autoencoder representations on pose-quality classification, even at sparsity levels where reconstruction is nearly perfect. Separately, pose filters built from the sparsest, and by the usual argument most interpretable, representations pass their own selection procedure but systematically fail to generalize to held-out data, while less aggressively sparse representations hold up reliably. Both failures trace to the same underlying cause: the magnitude-based feature selection that makes a representation sparse also discards information a downstream classifier needs and starves the feature-selection procedure of the examples it needs to estimate performance reliably. We report this as a cautionary result for anyone deploying SAE-based filters in biological pipelines: reconstruction fidelity is not a proxy for either predictive power or selection reliability, and both must be checked directly, at every sparsity level under consideration, before deployment.

1 Introduction

Predicting how a ligand binds a flexible protein target, the induced fit docking (IFD) problem [Sherman et al., 2006], is one of computational biochemistry's hardest problems: a ligand and a flexible receptor must be co-optimized against each other, and induced-fit effects that rigid-receptor docking ignores routinely make or break a pose's accuracy [Miller et al., 2021]. Interpretability is not a purely academic concern here: understanding *why* a model scores a pose highly is treated as being as important as the score itself for trustworthy deployment in drug discovery [Dhakal et al., 2022, Sim et al., 2025, Qadri et al., 2025]. Structure-informed, physics-grounded pipelines are an active area partly for this reason—they promise more interpretable, more generalizable predictions than end-to-end sequence-to-structure models [Wu et al., 2025, Lam and Katritch, 2025, Jiang et al., 2026].

Sparse autoencoders (SAEs) decompose opaque neural representations into interpretable, monosemantic features [Templeton et al., 2024, Gao et al., 2024, Bricken et al., 2023]. Applied to protein language models, they have already recovered biologically meaningful structure—secondary structure, binding sites, evolutionary conservation [Adams et al., 2025, Simon and Zou, 2025]—and the same approach has since been extended to structure-generation and protein-folding representations [Zarzecki et al., 2025, Parsan et al., 2025, Yu and Olson, 2026]. This growing body of positive results

Submitted to the I Can't Believe It's Not Better (ICBINB): Failure Modes of AI in Biology workshop at NeurIPS 2026.

shares an implicit assumption: decomposition comes cheaply. An SAE with enough hidden units and a small enough sparsity constraint can, in principle, reconstruct the dense representation almost perfectly. Scaling studies back this up directly, reporting clean, predictable reconstruction-vs-sparsity tradeoff curves as autoencoder width and K grow [Gao et al., 2024]. Under this assumption, any information present in the dense representation should still be recoverable from the sparse one, and performance on a downstream task built from the sparse representation should be no worse than performance built directly from the dense latent, at least once reconstruction fidelity is reasonably high. That expectation is borne out in at least one recent case, where downstream linear probes fit in an SAE's sparse latent space matched or exceeded probes fit directly on the dense representation [Yu and Olson, 2026].

We tested this expectation in exactly the kind of concrete, practically motivated setting described above. We study the Friesner Group's IFD-ML, a physics-informed protein-ligand docking pipeline that combines induced-fit docking [Sherman et al., 2006, Miller et al., 2021, Li et al., 2011] with a VAE and diffusion-based conformational sampling, achieving strong performance in cross-docking benchmarks similar to Mansoor et al. [2024]. It generates thousands of candidate poses per complex, each represented by a 30-dimensional continuous VAE latent vector, and each requiring expensive downstream energy minimization regardless of quality [Li et al., 2011].

A natural, practically valuable use of an SAE trained on this latent space is a pre-refinement filter: use SAE-derived features, rather than the raw latent, to flag likely-poor poses before they consume refinement compute. This requires two things to hold: the SAE representation should support pose-quality classification at least as well as the raw latent, and a filter selected on held-out data should generalize reliably to a separate test set. We test both across a realistic sweep of sparsity levels and find that neither holds: (1) raw latents match or beat every trained SAE on classification, even at sparsity levels with near-perfect reconstruction; and (2) filters built from the sparsest, and by the field's standard most interpretable, representations pass their own selection procedure but fail to generalize to held-out data, while less aggressively sparse representations hold up reliably.

2 Methodology

We trained TopK SAEs [Adams et al., 2025] on 435,069 docking poses across 36 protein-ligand systems, split by case (not pose) into $\sim$70/15/15 train/validation/test to prevent leakage. Each SAE uses a $4\times$ expansion (30D input $\rightarrow$ 120D hidden $\rightarrow$ 30D reconstruction); the TopK operation keeps only the K largest hidden activations per example and zeros the rest, with reconstruction loss as the sole training objective. The pre-encoder bias is initialized to the training-set mean of normalized activations and subtracted before encoding, following Simon et al. [2026]'s dead-feature mitigation. We swept $K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds per K, selecting the best run per K by validation loss.

We evaluated two downstream tasks on the resulting representations, both on a held-out test set never touched during training or selection:

Classification. Logistic regression classifiers (class-balanced) were fit to predict bad-pose vs. good-pose (RMSD $\leq$ 4Å native-like, following Shim et al. [2022]) from (a) the 30D normalized raw VAE latent (L2 penalty) and (b) the 120D SAE activations for each trained K (L1 penalty). We report test-set auPR, our primary metric given the 63.4% bad-pose base rate.

Filtering. For each K, we built a pre-refinement filter using a three-way train/select/test discipline: per-feature F1-maximizing activation thresholds and precision-based candidate ranking (requiring recall >0.02 to exclude degenerate picks) were fit on a pooled train+validation selection set; the top 16 candidates were then searched over all subset unions, on that same selection set, for the union maximizing recall subject to retaining $\geq$80% of native-like (good) poses. The winning union was applied exactly once to test. Here we report what happens when this procedure is applied uniformly across the full K sweep, rather than restricting to the small number of K values a practitioner would ultimately consider deploying.

3 Results and Discussion

3.1 Sparsity provides no consistent classification lift

Table 1 reports auPR and FVE across the full K sweep. Raw dense latents (auPR 0.770 at the 4Å threshold) beat every trained SAE K at this threshold, including K=30, whose FVE reaches 100%—near-perfect reconstruction does not translate into matching, let alone exceeding, the raw latent's classification performance. FVE grows essentially monotonically with K; auPR does not track it at all: K=6 (FVE 49.5%) modestly *outperforms* K=15 (FVE 90.0%) on 4Å auPR, and no K tested closes the gap to the raw latent.

Table 1: auPR and FVE, raw latents vs. SAE K (test set, full sweep).

Rep.	auPR (4Å)	auPR (2Å)	FVE	Dead %
Raw latents	0.770	0.855	—	—
SAE K=1	0.660	0.838	−0.2%	80.8%
SAE K=3	0.709	0.858	33.7%	2.5%
SAE K=6	0.737	0.867	49.5%	0.8%
SAE K=8	0.719	0.821	62.0%	1.7%
SAE K=15	0.750	0.846	90.0%	25.8%
SAE K=20	0.749	0.846	98.0%	30.8%
SAE K=30	0.742	0.810	100.0%	16.7%

At the 2Å threshold two SAE K's (3 and 6) edge past the raw latent by a small margin (0.858 and 0.867 vs. 0.855), but this is not a robust win for sparsity: it is not the K with the highest FVE, does not replicate at the 4Å threshold, and—as Section 3.2 shows—K=3 is precisely the regime where downstream filters built from this representation fail to generalize. The classification signal, in short, does not scale with reconstruction fidelity in any direction we can exploit.

This traces to a mechanism inherent to TopK sparsity itself, not to any particular K: the class-discriminative direction for "is this pose good or bad" need not coincide with the top-K largest-magnitude activations for a given example. TopK hard-thresholding is a magnitude-based, not relevance-based, selection rule [Gao et al., 2024]—it has no mechanism to preferentially retain the specific latent directions a downstream classifier would find useful, only the directions that happen to be locally largest. High reconstruction fidelity (FVE) certifies that the *representation as a whole* recovers the dense latent well on average; it does not certify that any single, small, per-example active subset preserves the specific low-variance directions a linear classifier depends on. This is consistent with K=30's near-perfect 100% FVE still trailing the raw latent on auPR: even when almost nothing is lost on reconstruction, something is still being lost on classification, because the two objectives are not the same objective.

3.2 Low-K filters overfit their own selection procedure

Table 2 reports, for every trained K, the good-pose-retention rate (good_kept) achieved by the winning filter combo on the selection set versus on held-out test, at the 4Å threshold. Every combo clears the 80% bar on the selection set, by construction of the search. But that guarantee does not carry over to test: $K \in \{1, 3, 6\}$ all fall short (75.6%, 77.5%, and a borderline 79.9% respectively), while $K \geq 8$ holds up reliably (86.1–95.9%). This is not an isolated failure of one unlucky K—it is systematic across every K below 8 that we trained, and the threshold sits precisely between the low-FVE, low-dead-feature regime and the higher-FVE regime where features begin dying at non-trivial rates (Table 1).

As a control, we ran the identical selection procedure on 500 random unit directions and all 60 signed raw latent axes in place of SAE features. Neither baseline family reached the 80% good_kept bar on test at all (0 of 65,535 candidate-union combinations; each family's closest achievable operating point is reported in Appendix A), so the low-K SAE filters are not simply behaving like unstructured linear directions—they are learning something, just not something the selection procedure can certify reliably at low K.

This failure, too, traces back to TopK sparsity itself: at low K, each pose activates only a handful of the 120 candidate features. Per-feature precision on the pooled train+validation selection set is therefore estimated from a small, noisy sample of firing poses for any single low-K feature, and the

Table 2: Selection-set vs. test-set good_kept for the winning filter combo at each trained K (4Å).

K	Selection good_kept	Test good_kept	Test precision	Test recall
1	82.4%	75.6%	0.575	0.191
3	85.2%	77.5%	0.698	0.301
6	82.8%	79.9%	0.754	0.355
8	83.6%	91.1%	0.803	0.210
15	80.8%	95.9%	0.879	0.171
20	84.9%	86.1%	0.621	0.132
30	82.2%	94.7%	0.817	0.137

combo-search procedure—which searches up to 65,535 subset unions of the top 16 candidates for the recall-maximizing combination subject to a retention constraint—has many degrees of freedom with which to fit whatever precision pattern that sampling noise happens to produce on the selection set. This is overfitting via researcher degrees of freedom, except the "researcher" here is an automated combinatorial search rather than a human choosing among a handful of options by hand, which makes it substantially easier to miss without an explicit, disciplined train/select/test split of the kind used here. Higher K mitigates this only incidentally: more active features per pose gives each candidate's precision estimate a larger, more stable firing sample, not because higher K is more interpretable—by the field's own standard, it is less so.

Both failures, in short, share a root cause: TopK sparsity forces each pose's 120-dimensional representation down to its K largest activations, and K is small precisely where the representation is most aggressively sparse and thus most interpretable by the usual argument for using SAEs at all [Templeton et al., 2024, Bricken et al., 2023]. The K values most worth reporting as interpretable, monosemantic decompositions are exactly the K values whose downstream filters are least trustworthy without an explicit generalization check.

4 Conclusion

The practical upshot is a genuine tension for anyone deploying SAE-based filters in a scientific pipeline: the K values most worth reporting as interpretable, monosemantic decompositions are exactly the K values whose downstream filters are least trustworthy without an explicit, held-out generalization check, and whose classification performance is not guaranteed to match the raw representation even when reconstruction fidelity looks excellent. Neither failure is visible from reconstruction loss or FVE alone; both only appear once the downstream task is evaluated on genuinely held-out data. We report this as a concrete argument for treating FVE as, at best, a necessary and not sufficient diagnostic before deploying an SAE-based filter, and for always including a raw-latent and a held-out generalization baseline alongside any reported SAE result, at every K under consideration rather than only the one ultimately deployed.

We do not read this as a contradiction of the positive results the protein-language-model SAE literature has reported [Adams et al., 2025, Simon and Zou, 2025, Zarzecki et al., 2025, Parsan et al., 2025, Yu and Olson, 2026]—those results are about the *qualitative* interpretability of individual features, established by inspecting what a feature fires on, and nothing here disputes that TopK SAEs can find monosemantic, human-legible structure in a biological latent space. What our result complicates is the further, usually implicit, step from "these features are interpretable" to "these features are safe to use in place of the dense representation for a downstream task." That step is a separate empirical claim, and in our setting it is false at exactly the sparsity levels the interpretability claim is strongest at. Reporting an SAE's qualitative feature interpretability and its downstream reliability are two different experiments, and a pipeline that only runs the first risks deploying a filter whose quantitative behavior was never actually checked. Given how cheaply the second experiment can be added—it is the same train/select/test split already needed for any responsible use of a data-derived filter—we think it should be standard practice alongside, not instead of, feature-level interpretability analysis.

References

Etowah Adams, Liam Bai, Minji Lee, Yiyang Yu, and Mohammed AlQuraishi. From mechanistic interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders on protein language models. *bioRxiv*, 2025.

T. Bricken et al. Towards monosemanticity: Decomposing language models with sparse autoencoders. *Transformer Circuits Thread*, 2023. URL `https://transformer-circuits.pub/2023/monosemantic-features/index.html`.

A. Dhakal, C. McKay, J. J. Tanner, and J. Cheng. Artificial intelligence in the prediction of protein–ligand interactions: Recent advances and future directions. *Briefings in Bioinformatics*, 23 (1):bbab476, 2022.

L. Gao, T. D. la Tour, H. Tillman, G. Goh, et al. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.

J. Jiang, D. Li, G. Wang, and G.-W. Wei. Recent advances in machine learning predictions of protein-ligand binding affinities. *Current Opinion in Structural Biology*, 96:103193, 2026.

J. H. Lam and V. Katritch. Navigating structure-based drug discovery with emerging innovations in physics- and knowledge-based approaches. *Npj Drug Discovery*, 2(1):29, 2025.

Jianing Li, Robert Abel, Kai Zhu, Yixiang Cao, Suwen Zhao, and Richard A. Friesner. The VSGB 2.0 model: A next generation energy model for high resolution protein structure modeling. *Proteins: Structure, Function, and Bioinformatics*, 79(10):2794–2812, 2011.

S. Mansoor, M. Baek, H. Park, G. R. Lee, and D. Baker. Protein ensemble generation through variational autoencoder latent space sampling. *Journal of Chemical Theory and Computation*, 20 (7):2689–2695, 2024.

E. B. Miller, R. B. Murphy, D. Sindhikara, K. W. Borrelli, et al. Reliable and accurate solution to the induced fit docking problem for protein–ligand binding. *Journal of Chemical Theory and Computation*, 17(4):2630–2639, 2021.

Nithin Parsan, David J. Yang, and John J. Yang. Towards interpretable protein structure prediction with sparse autoencoders. *arXiv preprint arXiv:2503.08764*, 2025.

Y. A. Qadri, S. Shaikh, K. Ahmad, I. Choi, et al. Explainable artificial intelligence: A perspective on drug discovery. *Pharmaceutics*, 17(9):1119, 2025.

Woody Sherman, Tyler Day, Matthew P. Jacobson, Richard A. Friesner, and Ramy Farid. Novel procedure for modeling ligand/receptor induced fit effects. *Journal of Medicinal Chemistry*, 49(2): 534–553, 2006.

H. Shim, H. Kim, J. E. Allen, and H. Wulff. Pose classification using three-dimensional atomic structure-based neural networks applied to ion channel–ligand docking. *Journal of Chemical Information and Modeling*, 62(10):2301–2315, 2022.

J. Sim, D. Kim, B. Kim, J. Choi, and J. Lee. Recent advances in AI-driven protein-ligand interaction predictions. *Current Opinion in Structural Biology*, 92:103020, 2025.

E. Simon and J. Zou. InterPLM: discovering interpretable features in protein language models via sparse autoencoders. *Nature Methods*, 22:2107–2117, 2025.

Elana Simon, Etowah Adams, and James Zou. On the relationship between activation outliers and feature death in sparse autoencoders. *arXiv preprint arXiv:2605.31518*, 2026.

A. Templeton, T. Conerly, J. Marcus, J. Lindsey, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL `https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html`.

M.-H. Wu, Z. Xie, and D. Zhi. A folding-docking-affinity framework for protein-ligand binding affinity prediction. *Communications Chemistry*, 8(1):108, 2025.

Michael Yu and Matthew L. Olson. VFUSE: Virulent feature understanding with sparse autoencoders. *arXiv preprint arXiv:2606.10080*, 2026.

W. Zarzecki, P. Szymczak, E. Szczurek, and K. Deja. FoldSAE: Learning to steer protein folding through sparse representations. *arXiv preprint arXiv:2511.22519*, 2025.

A Raw-latent-geometry control: full baseline results

To confirm that the generalization gap in Section 3.2 reflects something the SAE decomposition is doing, rather than an artifact of the selection-and-filter procedure itself, we applied the identical train/select/test discipline (Section 2, Filtering) to two raw-latent-geometry baselines in place of SAE features: 500 random unit directions in the 30D VAE latent space, and all 60 signed raw latent axes (both positive and negative directions of each of the 30 dimensions). Table 3 reports each baseline family's closest achievable operating point at both RMSD thresholds used in the underlying dataset.

Table 3: Raw-latent-geometry baselines' closest achievable operating point under the identical selection discipline (test set). No combination from either family reaches the 80% good_kept target used to select SAE filter combos.

Threshold	Source	% Flagged	Precision	Recall	good_kept
4Å	Random directions (best)	33.7	0.605	0.321	63.6%
4Å	Raw latent axes (best)	27.8	0.727	0.319	79.2%
2Å	Random directions (best)	33.7	0.824	0.325	59.7%
2Å	Raw latent axes (best)	27.8	0.857	0.280	72.8%

Both baseline families come reasonably close to the SAE filters' precision at a comparable flag rate—raw latent axes even edge out low-K SAE precision at 2Å (0.857 vs. SAE K=3's 0.791)—but neither ever clears the 80% good_kept bar across all 65,535 candidate-union combinations searched per family, at either threshold. This rules out the simplest alternative explanation for Section 3.2's low-K generalization failure: it is not that *any* linear direction in this latent space produces a selection-set-vs-test gap of similar severity. The SAE decomposition is finding structure that plain latent geometry does not; the failure specific to low K is a property of how TopK sparsity interacts with the candidate-selection procedure, not a property of the latent space itself.